

Reg. No.:

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TERM END EXAMINATIONS (TEE) – Dec 2024 – Jan 2025

Programme	: B. Tech.	Semester	: Interim Semester 2024-25
Course Title	: Computer Architecture and Organization	Course Code	: CSE2003
Date	: 03-01-2024	Slot	: C11+C12+C13
Time	: 1½ hours	Max. Marks	: 50

Answer all the Questions

Q.No.

Question Description

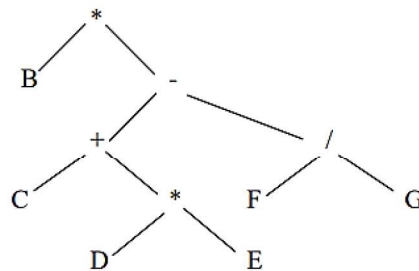
Marks

PART - A (30 Marks)

- 1 (a) Draw the Common Bus Organization of a simple 8085 Computer. 10

OR

- (b) The arithmetic expression represented below. Solve the expression and write it as 0, 1, 2 and 3 address instruction format.



- 2 (a) Given that Cache is 8-way set-associative. Find the following information using fig: 10
- Main Memory size.
 - Cache size
 - Block size.
 - Number of sets.
 - Total cache blocks

Tag (18 bits)	Set Index (6 bits)	Block Offset (6 bits)
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OR

(b) Consider the following instruction sequence:

I₁: ADD R₁, R₂, R₃
 I₂: SUB R₄, R₁, R₅
 I₃: ADD R₁, R₆, R₇
 I₄: MUL R₈, R₁, R₉

10

Identify any RAW, WAW or WAR hazards in the sequence. Explain how each hazard affects pipeline execution.

3 (a) Assume a processor has only one memory unit and the following sequence of instructions is executed:

I₁: LW R₁, 0(R₂)
 I₂: LW R₃, 4(R₂)
 I₃: ADD R₄, R₁, R₃

10

What type of hazard might occur between I₁ and I₂? How can structural hazards be resolved in this pipeline?

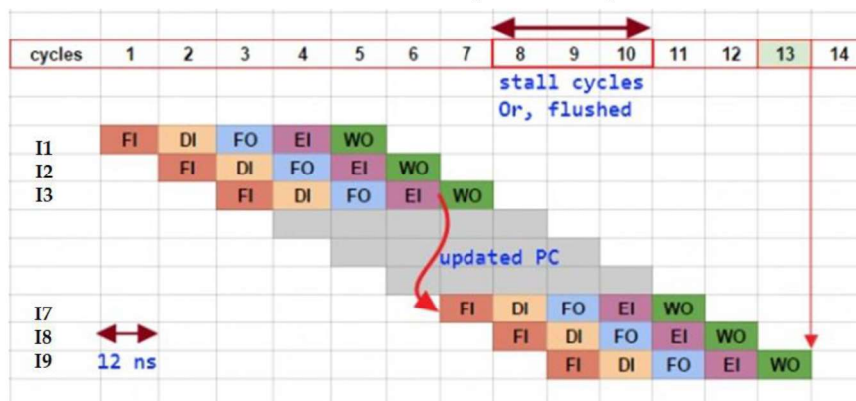
OR

(b) Explain with the help of diagram how DMA is beneficial in bulk data transfers? Discuss five use-cases in which components often use DMA.

10

PART - B (20 Marks)

4 (a)



10

Determine mathematically the total execution time for the given instructions, both with and without branching instruction.

5 Explain the difference between the concepts depicted in figures 1 and 2.

Core 1: | Thread A (Task 1) |
 Core 2: | Thread B (Task 2) |
 Core 3: | Thread C (Task 3) |
 Core 4: | Thread D (Task 4) |

Fig 1

10

Thread A: | Instr 1 | Instr 2 | Instr 3 | Instr 4 |
 _____|_____|_____|_____| (overlapped execution)

Fig 2